Interpretable EEG-to-fMRI prediction across people and recording sites

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Lagged EEG power outperforms NeuroBOLT and supports an arousal-monitor prototype.

Background and research question

EEG measures electrical activity at the scalp; fMRI BOLD reflects slower changes in blood oxygenation. Predicting BOLD from EEG could help monitor fluctuations in brain state.

NeuroBOLT [1] combines spectral features with a pretrained transformer. Can an interpretable EEG model predict unseen people and generalise across sites?

Methodology

VU: 22 people, 29 eyes-closed scans (~20 min).
External ds006040 [2]: 28 people, 56 runs (~6 min), eyes closed and open; a different scanner and EEG cap.

1 Hz log power (26 channels / 1-45 Hz) → **24 s history** (12 lags of 2 s) → **Ridge regression** (Regional + global BOLD)

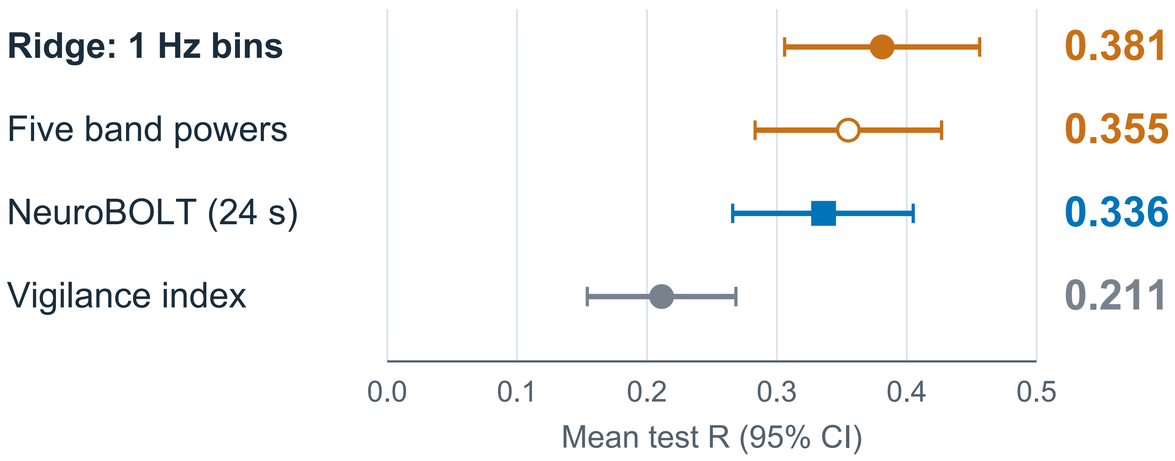
Five folds grouped by person; separate validation people for tuning.
Frozen-model test at the new site. Mean *R* and 95% CIs over people.

Results

Prediction, interpretation and an independent-site application

1. Prediction in unseen people

VU cohort | 7 targets | 22 held-out people

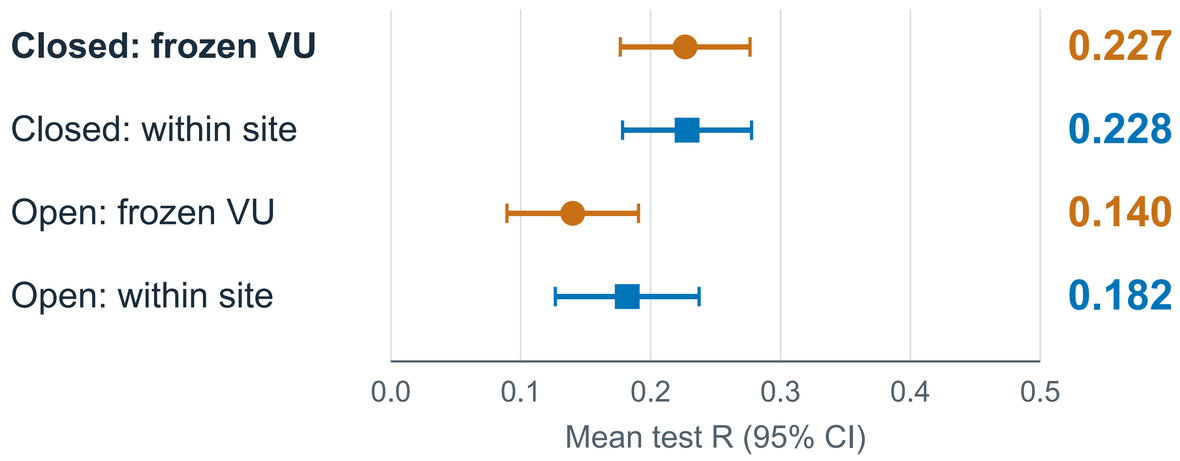


Ridge led NeuroBOLT in **19/22 people**:
ΔR = +0.045 (95% CI +0.023 to +0.067).

Seven targets: six DiFuMo regions + global BOLD. NeuroBOLT: shared 24 s model, three seeds. Vigilance index: alpha/(delta + theta) power.

2. Transfer to an independent site

ds006040 | 7 targets | 28 new people



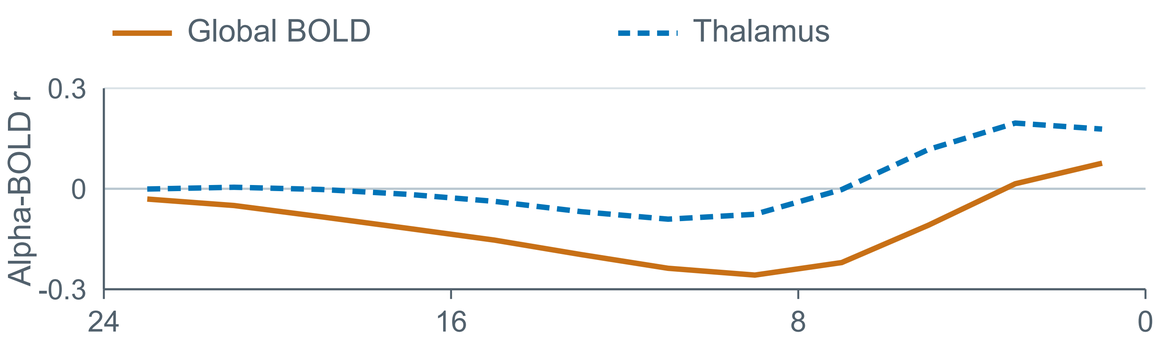
Eyes closed: similar means for frozen and local models. Paired ΔR (local - frozen) = +0.001 (95% CI -0.017 to +0.020).

Published NeuroBOLT reached R = 0.180 with eyes closed. Eyes-open rest was harder for the frozen ridge model.

3. What carries the prediction?

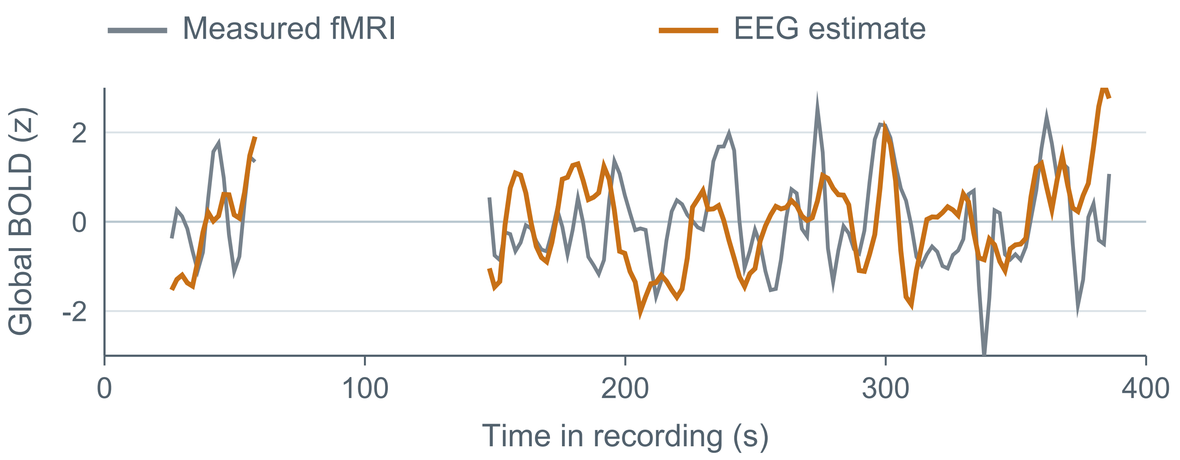
0.343 → 0.144

Mean R across the same 64 regions, before and after global-signal regression (GSR).



Alpha power couples negatively with global BOLD around 9 s earlier; the thalamus has a positive component around 3 s. This supports a brain-state interpretation, with regional differences.

4. Prototype: EEG arousal monitor



Example: sub-008_EC (r = 0.28), closest to median eyes-closed accuracy. Gap = excluded EEG windows.

r = 0.263

~3 ms / volume

Streaming replay: 28 eyes-closed runs.
95% CI 0.205 to 0.321; one CPU core.
EEG estimates global BOLD; fMRI assesses accuracy.

Conclusion

Penalised linear models provide a strong baseline for EEG-to-BOLD prediction. Much of the predictable signal follows brain-wide state fluctuations. Transfer worked on one independent site; the model supports an arousal-monitor prototype.

Limitations and further research

Test region-specific prediction with respiration and motion controls at more sites. Validate arousal flags against pupil measurements or sleep staging. These are observational results; streaming used replayed, cleaned EEG. Prospective use remains untested.

References and resources

[1] Li et al. NeuroBOLT: Resting-state EEG-to-fMRI synthesis with multi-dimensional feature mapping. NeurIPS, 2024.
[2] Cha et al. Sustained attention task (gradCPT) dataset using simultaneous EEG-fMRI and DTI. Scientific Data, 2026. doi:10.1038/s41597-026-06616-6. OpenNeuro: ds006040.
Analysis code, model weights and full reports: github.com/seamus-programming-stuff/NeuroBOLT

Code + reports